(iii) $v = f\lambda = 1060 \times 0.32 = 339\ \text{m s}^{-1}$

5 **(a)** e.m.f. = 3.0 V.

(b)
$$V_T = E - Ir$$
$$2.20 = 3.00 - 1.60 \times r$$
$$r = \frac{3.00 - 2.20}{1.60} = 0.50\ \Omega$$

(c) **(i)** power $= IV = 1.60 \times 2.20 = 3.52$ W

(ii)
$$\text{efficiency} = \frac{\text{power delivered to lamp}}{\text{total power}} = \frac{IV}{IE} = \frac{V}{E}$$
$$= \frac{2.20}{3.00}$$
$$= 73.3\%$$

6 **(a)** The magnetic flux density of a magnetic field is numerically equal to the force per unit